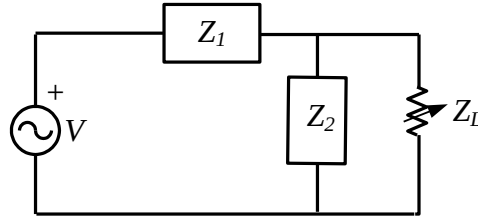


Name:

R. No:

Section:

1. In the figure Z_L consists of a variable resistance and a fixed capacitive reactance of 4Ω . If $V = 50 \angle 0^\circ \text{V}$, $Z_1 = Z_2 = 10 + j24 \Omega$, find the value of Z_L for maximum power transfer. Also determine the maximum power. [3]



2. Obtain $X(t)$ by solving the following state equations. Assume the initial conditions of the state variables are zero. The input is a unit step function. [6]

$$\dot{x}_1 = x_1 - 2x_2 + u$$

$$\dot{x}_2 = 3x_1 - 4x_2 + 3u$$

3. The voltage transfer function of a linear electrical circuit is $G(s) = \frac{2}{s^2 + 2s + 5}$. If the steady-state response of the circuit is $2 + 4 \sin 2t$ Volts, determine the input voltage. [6]